

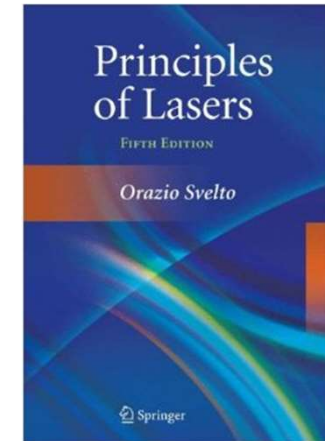


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch2-5 Open Cavity & Gaussian Beam



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Chapter 2: Open Cavity & Gaussian Beam

❖ 2.9 The Gaussian beam

- Expression of Gaussian beam
- The transmission rule in free space
- Characteristic Parameter of Gaussian beam's fundamental mode

❖ 2.10 Transformation Rule of Gaussian beam

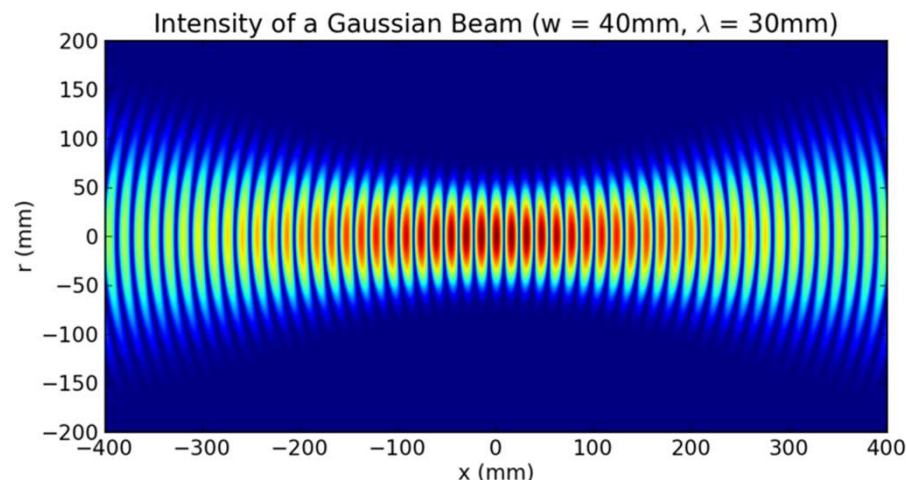
- The ABCD law
- The application of ABCD law

2.9 The Gaussian Beam

❖ Expression of the Gaussian Beam

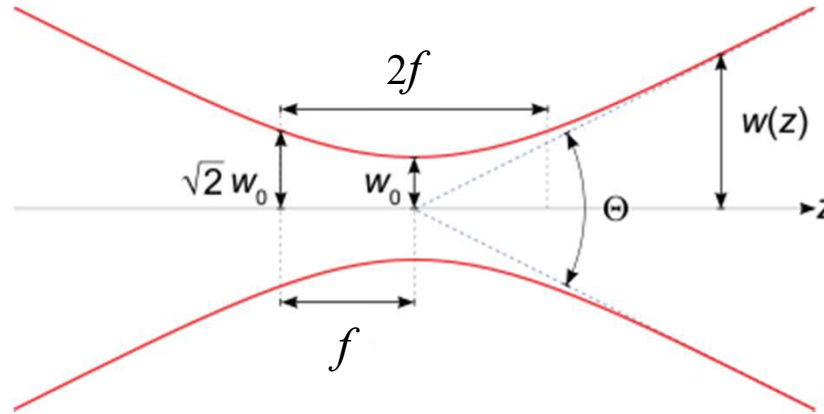
The laser beam from a laser device is not the homogeneous plane wave, and is not the homogeneous spherical wave, but Gaussian beam, that is spherical Gaussian beam whose amplitude and phase is changing.

A Gaussian beam is a beam of electromagnetic radiation whose transverse electric field and intensity (irradiance) distributions are well approximated by Gaussian functions. Many lasers emit beams that approximate a Gaussian profile, in which case the laser is said to operate on the **fundamental transverse mode**, or “**TEM₀₀ mode**” of the laser's optical resonator.



A 5 mW green laser pointer beam profile, showing the TEM₀₀ profile





Fundamental mode Gaussian beam, propagating along z direction

$$\Psi_{00}(x, y, z) = \frac{c}{w(z)} \exp\left[-\frac{r^2}{w^2(z)}\right] \exp\left\{-i\left[k\left(z + \frac{r^2}{2R}\right) - \arctg \frac{z}{f}\right]\right\}$$

Where c is constant, $r^2 = x^2 + y^2$, $k = 2\pi/\lambda$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{f}\right)^2} \quad R = R(z) = z \left[1 + \left(\frac{z}{f}\right)^2\right] = f \left(\frac{z}{f} + \frac{f}{z}\right)$$

$$f = \pi w_0^2 / \lambda, \quad w_0 = \sqrt{f \lambda / \pi}$$

w_0 Minimum spot size of the Gaussian beam (beam waist)
 f Confocal parameter of the Gaussian beam

❖ The transmission rule in free space

(1) The amplitude of the fundamental mode field cross-section follows the Gaussian distribution. The beam radius is equal to

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{f}\right)^2} = w_0 \sqrt{1 + \left(\frac{\lambda z}{\pi w_0^2}\right)^2}$$

(2) The phase factor: phase shift characteristics of fundamental mode

$$\Phi_{00}(x, y, z) = k \left(\underbrace{z}_{\text{geometric phase shift}} + \underbrace{\frac{r^2}{2R}}_{\text{phase shift related to } x \text{ and } y} \right) - \underbrace{\arctg \frac{z}{f}}_{\text{additional phase shift beside } z}$$

$$R(z) = z + \frac{f^2}{z} = z \left[1 + \left(\frac{\pi w_0^2}{\lambda z} \right)^2 \right] = \begin{cases} R(z) \rightarrow \infty & z = 0 \\ |R(z)| \approx |z| \rightarrow \infty & z = \pm \infty \\ |R(z)| = 2f & z = \pm f \\ R(z) > 2f & 0 < z < f \\ z < R(z) < z + f & z > f \end{cases}$$

The wave-front is flat or planar shape at $z=0$, corresponding to an infinite curvature or $R(z)=\infty$. As the beam propagates, however, the wave-front gradually become curved, and the $R(z)$ drops rather rapidly to finite values.

For distances well beyond the **Rayleigh Range f** the curvature then increases again as $R(z)=z$, i.e., the Gaussian beam becomes essentially like a spherical wave centered at the beam waist ($z=0$). Physically, it means that the centers of curvature of the wave-fronts located at various z positions have the same center ($z=0$). The curvature increases from $R(z)$ where $z=f$ to ∞ , when the wave-front itself moves as z approach $+\infty$.

(3) The far-field divergence angle

$$\theta_0 = \lim_{z \rightarrow \infty} \frac{2w(z)}{z} = 2 \frac{\lambda}{\pi w_0} = 0.6367 \frac{\lambda}{w_0} = 2 \sqrt{\frac{\lambda}{\pi f}} = 1.128 \sqrt{\frac{\lambda}{f}}$$

Gaussian beam can be approximately regarded as **inhomogeneous spherical wave** around its axis transmission. The center of curvature changes constantly with the transmission process. The amplitude and intensity follow **Gaussian distribution** within the cross section, and their **equiphase surface was spherical**.

❖ Characteristic Parameter of Gaussian beam's fundamental mode

(1) f - z parameters: Using w_0 (or f) and z to describe Gaussian beam

(2) w - R parameters: Using $w(z)$ and $R(z)$ to describe Gaussian beam

$$\left. \begin{aligned} w(z) &= w_0 \sqrt{1 + \left(\frac{z}{f}\right)^2} \\ R(z) &= z + \frac{f^2}{z} \end{aligned} \right\} \quad \boxed{\begin{aligned} w_0 &= w(z) \left[1 + \left(\frac{\pi w^2(z)}{\lambda R(z)} \right)^2 \right]^{-1/2} \\ z &= R(z) \left[1 + \left(\frac{\lambda R(z)}{\pi w^2(z)} \right)^2 \right]^{-1} \end{aligned}}$$

(3) q parameter: complex curvature

$$\begin{aligned} \Psi_{00}(x, y, z) &= \frac{c}{w(z)} \exp\left[-\frac{r^2}{w^2(z)}\right] \exp\left\{-i\left[k\left(z + \frac{r^2}{2R(z)}\right) - \arctg \frac{z}{f}\right]\right\} \\ &= \frac{c}{w(z)} \exp\left\{-ik \frac{r^2}{2} \left[\frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}\right]\right\} \exp\left[-i\left(kz - \arctg \frac{z}{f}\right)\right] \end{aligned}$$

$$\Psi_{00}(x, y, z) = \frac{c}{w(z)} \exp \left\{ -ik \frac{r^2}{2} \left[\frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)} \right] \right\} \exp \left[-i \left(kz - \arctg \frac{z}{f} \right) \right]$$

Introduce a new q parameter

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}$$

$$\Psi_{00}(x, y, z) = \frac{c}{w(z)} \exp \left(-ik \frac{r^2}{2} \frac{1}{q} \right) \exp \left[-i \left(kz - \arctg \frac{z}{f} \right) \right]$$

Once we know the q parameter,
we can obtain $R(z)$ and $w(z)$

$$\begin{cases} \frac{1}{R(z)} = \operatorname{Re} \left\{ \frac{1}{q(z)} \right\} \\ \frac{1}{w^2(z)} = -\frac{\pi}{\lambda} \operatorname{Im} \left\{ \frac{1}{q(z)} \right\} \end{cases}$$

Proof $q(z) = z + if$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{f}\right)^2} \quad R(z) = z + \frac{f^2}{z} \quad w_0 = \sqrt{\frac{\lambda f}{\pi}}$$

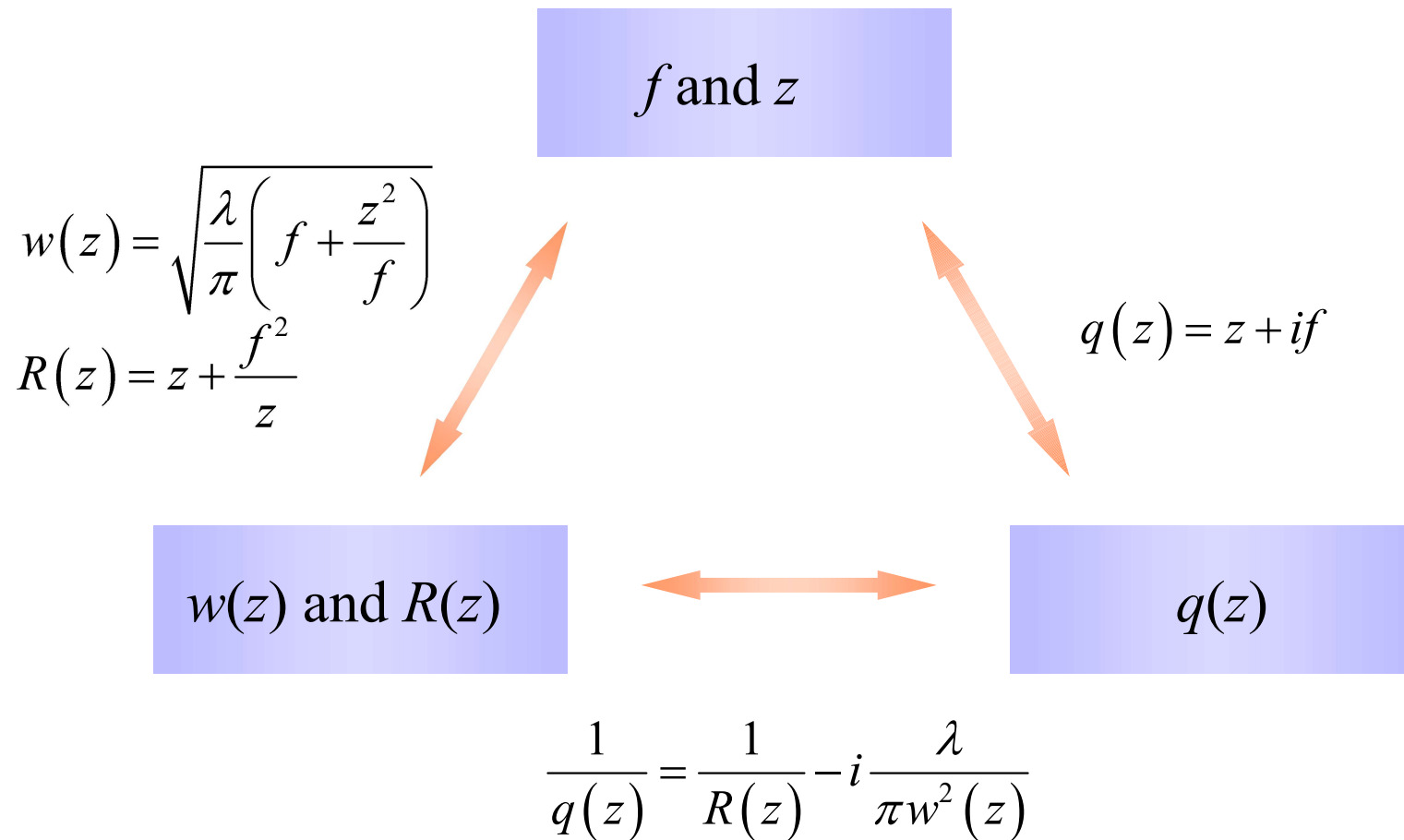
$$\begin{aligned} \frac{1}{q(z)} &= \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)} = \frac{z}{z^2 + f^2} - i \frac{\lambda}{\pi \frac{\lambda f}{\pi} \left(\frac{z^2 + f^2}{f^2} \right)} \\ &= \frac{z}{z^2 + f^2} - i \frac{f}{z^2 + f^2} = \frac{z - if}{z^2 + f^2} \end{aligned}$$

$$q(z) = \frac{z^2 + f^2}{z - if} = \frac{(z^2 + f^2)(z + if)}{(z - if)(z + if)} = z + if$$

Discussion: at the beam waist $z=0$

$$\frac{1}{q_0} = \frac{1}{q(0)} = \frac{1}{R(0)} - i \frac{\lambda}{\pi w^2(0)} \quad \xrightarrow{\quad} \quad q_0 = i \frac{\pi w^2(0)}{\lambda} = if$$

Three methods to describe Gaussian beam



Exercise 1: The wavelength of the Gaussian beam is $3.14 \mu\text{m}$, and the minimum spot size $w_0=1 \text{ mm}$. Write down the q parameter, the beam size w , and the curvature R of the equivalent phase surface at position $z=50 \text{ cm}$.

$$(1) \text{ } q \text{ parameter} \quad w_0 = \sqrt{\lambda f / \pi} \quad f = \frac{\pi w_0^2}{\lambda} = \frac{3.14 \times 10^{-6}}{3.14 \times 10^{-6}} = 1 \text{ m}$$

$$z = 0.5 \text{ m} \quad q = 0.5 + i(m)$$

(2) Beam size w and curvature R of the equivalent phase surface

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{f}\right)^2} = 1 \sqrt{1 + \left(\frac{0.5}{1}\right)^2} = 1.12 \text{ m}$$

$$R(z) = z + \frac{f^2}{z} = 0.5 + \frac{1^2}{0.5} = 2.5 \text{ m}$$

Exercise 2: At one specific z position, the beam size of the Gaussian beam is $w(z)=1 \text{ mm}$, and the curvature of the equivalent phase surface $R(z)=0.5 \text{ m}$. Write the (1) The q parameter of the Gaussian beam at this position. (2) The radius and position of the beam waist (wavelength $\lambda=3.14 \mu\text{m}$).

(1) q parameter $\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)} = \frac{1}{0.5} - i \frac{3.14 \times 10^{-6}}{3.14 \times (10^{-3})^2} = 2 - i$

$$q = \frac{1}{2 - i} = \frac{2 + i}{4 + 1} = \frac{2 + i}{5} = 0.4 + 0.2i(m)$$

(2) Position and radius w_0 of the beam waist

$$\begin{aligned} w(z) &= \sqrt{\frac{\lambda}{\pi} \left(f + \frac{z^2}{f} \right)} & f + \frac{z^2}{f} &= \frac{z^2 + f^2}{f} = 1 \\ R(z) &= z + \frac{f^2}{z} & z + \frac{f^2}{z} &= \frac{z^2 + f^2}{z} = 0.5 \end{aligned} \quad \left. \vphantom{\begin{aligned} w(z) &= \sqrt{\frac{\lambda}{\pi} \left(f + \frac{z^2}{f} \right)} \\ R(z) &= z + \frac{f^2}{z} \end{aligned}} \right\} z = 2f$$

$$f = 0.2m$$

$$z = 0.4m$$

$$w_0 = \sqrt{\frac{\lambda f}{\pi}} = \sqrt{\frac{3.14 \times 10^{-6} \times 0.2}{3.14}} = 0.447mm$$

Exercise 3: The wavelength of the Gaussian beam is $3.14 \mu\text{m}$. If the q parameter at one specific position is $q=1+i$ (m). (1) The radius of the beam waist and this specific position. (2) The beam size w and the curvature R of the equivalent phase surface at this position.

(1) The radius and position of the beam waist

$$z = 1\text{m} \quad f = 1\text{m}$$

$$w_0 = \sqrt{\frac{\lambda f}{\pi}} = \sqrt{\frac{3.14 \times 10^{-6} \times 1}{3.14}} = 1\text{mm}$$

(2) The beam size w and the curvature R of the equivalent phase surface

$$\frac{1}{q} = \frac{1}{1+i} = \frac{1-i}{2} = \frac{1}{2} - i \frac{1}{2} = \frac{1}{R} - i \frac{\lambda}{\pi w^2}$$

$$\frac{1}{2} = \frac{1}{R} \quad R = 2\text{m}$$

$$\frac{1}{2} = \frac{\lambda}{\pi w^2} \quad w = \sqrt{\frac{2\lambda}{\pi}} = \sqrt{\frac{2 \times 3.14 \times 10^{-6}}{3.14}} = 1.414\text{mm}$$

2.10 Transformation Rule of Gaussian Beam

❖ The $ABCD$ law

The transformation matrix of optical component is $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$

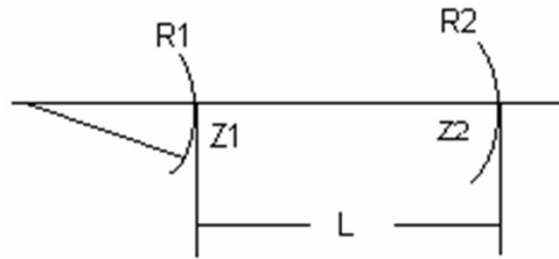
For Gaussian beam, if the radius of curvature is $R_1(z)$ and the parameter is $q_1(z)$, after passing through this optical component, the radius of curvature $R_2(z)$ and the $q_2(z)$ parameter can be described by

$$\begin{aligned} R_2 &= \frac{AR_1 + B}{CR_1 + D} & R_1 \text{ and } q_1 \text{ are the parameters before the optical component} \\ q_2 &= \frac{Aq_1 + B}{Cq_1 + D} & R_2 \text{ and } q_2 \text{ are the parameters after the optical component} \end{aligned}$$

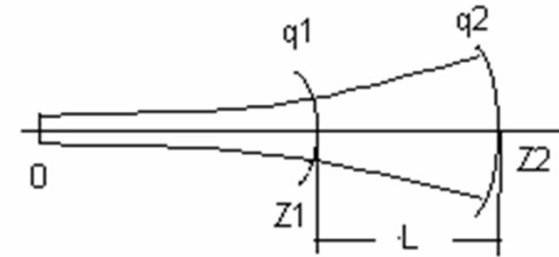
1) Free space

$$T = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

Matrix



Spherical wave



Gaussian beam

For R

$$R_2 = \frac{AR_1 + B}{CR_1 + D} = \frac{R_1 + L}{0R_1 + 1} = R_1 + L$$

$$\left. \begin{array}{l} R_1 = z_1, R_2 = z_2 \\ R_2 = R_1 + (z_2 - z_1) = R_1 + L \end{array} \right\} R_2 = R_1 + L$$

For q

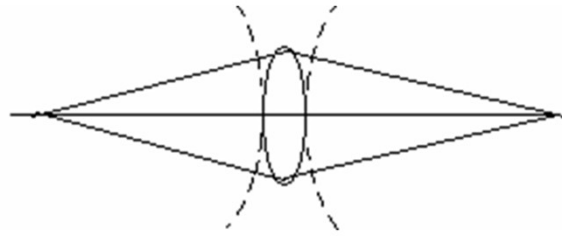
$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} = \frac{1 \times q_1 + L}{0 \times q_1 + 1} = q_1 + L$$

$$\left. \begin{array}{l} q_1 = z_1 + if \\ q_2 = z_2 + if \end{array} \right\} q_2 - q_1 = z_2 - z_1 = L$$

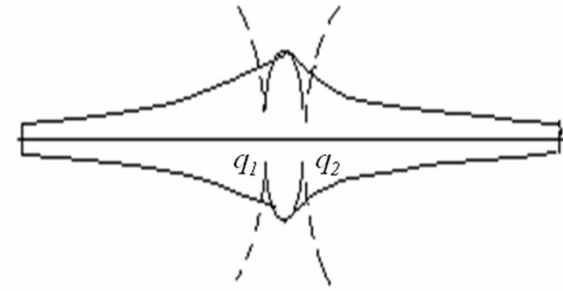
2) Lens

$$T = \begin{pmatrix} 1 & 0 \\ -\frac{1}{F} & 1 \end{pmatrix}$$

Matrix



Spherical wave



Gaussian beam

For R

$$R_2 = \frac{AR_1 + B}{CR_1 + D} = \frac{R_1 + 0}{-R_1/F + 1} = \frac{FR_1}{F - R_1}$$

$$\left. \begin{aligned} \frac{1}{R_1} + \frac{1}{-R_2} &= \frac{1}{F} \\ \frac{1}{R_2} &= \frac{1}{R_1} - \frac{1}{F} = \frac{F - R_1}{FR_1} \end{aligned} \right\} R_2 = \frac{FR_1}{F - R_1}$$

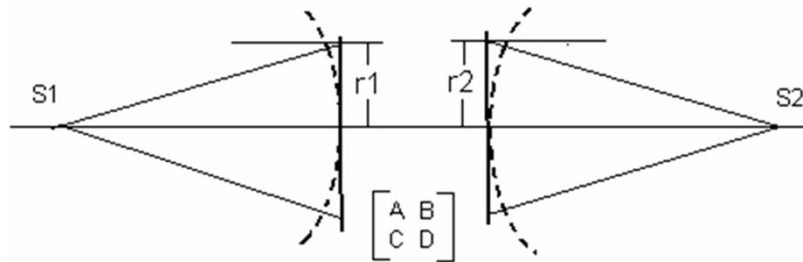
For q

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} = \frac{1 \times q_1 + 0}{-q_1/F + 1} = \frac{Fq_1}{F - q_1}$$

$$\left. \begin{aligned} \frac{1}{q_1} &= \frac{1}{R_1} - i \frac{\lambda}{\pi w_1^2} & \frac{1}{q_2} &= \frac{1}{R_2} - i \frac{\lambda}{\pi w_2^2} \\ \frac{1}{R_2} &= \frac{1}{R_1} - \frac{1}{F} & w_2 &= w_1 \end{aligned} \right\}$$

$$\frac{1}{q_2} - \frac{1}{q_1} = \frac{1}{R_2} - \frac{1}{R_1} = -\frac{1}{F} \Rightarrow q_2 = \frac{Fq_1}{F - q_1}$$

3) Any optical element



$$\begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix}$$

Spherical wave

$$\begin{aligned} r_2 &= Ar_1 + B\theta_1 \\ \theta_2 &= Cr_1 + D\theta_1 \end{aligned} \quad \xrightarrow{r_2 \approx R_2\theta_2 \quad r_1 \approx R_1\theta_1} \quad R_2 = \frac{r_2}{\theta_2} = \frac{AR_1 + B}{CR_1 + D}$$

Gaussian beam: The change of q parameter through optical system is the same as transformation of spherical wave R

Free space	$R_2 = R_1 + L$	$q_2 = q_1 + L$		$q_2 = \frac{Aq_1 + B}{Cq_1 + D}$
Lens	$\frac{1}{R_2} = \frac{1}{R_1} - \frac{1}{F}$	$\frac{1}{q_2} = \frac{1}{q_1} - \frac{1}{F}$		
	Spherical wave	Gaussian beam		The <i>ABCD</i> law

The powerful *ABCD* law enables us to trace the Gaussian beam parameter $q(z)$ through a complicated sequence of lens-like elements. The radius of curvature $R(z)$ and the spot size $w(z)$ at any plane z can be calculated through the use of the following expression.

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}$$

❖ The application of $ABCD$ law

If we know w_0 , l and F

After passing the lens, what is w_0' and R_0'

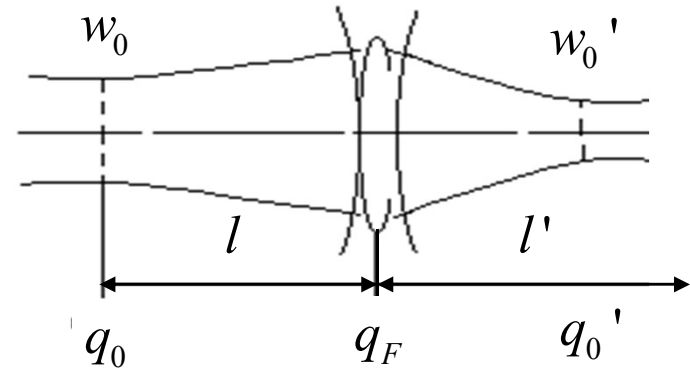
$$q_0 = if = i \frac{\pi w_0^2}{\lambda} \quad q_F = q_0' - l' = i \frac{\pi w_0'^2}{\lambda} - l'$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{F} & 1 \end{pmatrix} \begin{pmatrix} 1 & l \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & l \\ -\frac{1}{F} & 1 - \frac{l}{F} \end{pmatrix}$$

$$q_F = \frac{Aq_0 + B}{Cq_0 + D} = \frac{i \frac{\pi w_0^2}{\lambda} + l}{-\frac{1}{F} i \frac{\pi w_0^2}{\lambda} + \left(1 - \frac{l}{F}\right)} = i \frac{\pi w_0'^2}{\lambda} - l'$$

$$\frac{w_0'}{w_0} = \frac{F}{\sqrt{(l-F)^2 + f^2}}$$

$$l' = F + \frac{(l-F)F^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2} \quad w_0'^2 = \frac{F^2 w_0^2}{(l-F)^2 + \left(\frac{\pi w_0^2}{\lambda}\right)^2}$$



Discussion: imaging of Gaussian beam and geometric optics

1) For $l \gg F \Rightarrow (l - F)^2 \gg f^2$

$$l' = F + \frac{(l - F)F^2}{(l - F)^2 + f^2} \approx F + \frac{F^2}{l - F} = \frac{lF}{l - F} \quad \Rightarrow \quad \boxed{\frac{1}{l'} + \frac{1}{l} = \frac{1}{F}}$$

2) When $l = F$

$$l' = F + \frac{(l - F)F^2}{(l - F)^2 + f^2} = F$$

geometric optics

$$l' = \infty$$

$$w_0'^2 = \frac{w_0^2 F^2}{(F - l)^2 + f^2} \xrightarrow{l = F} w_0'^2 = \frac{w_0^2 F^2}{f^2}$$

3) When $l < F \quad 0 < l' < F$

geometric optics $l < F$ no image